

LINDENBERG, Germany

WMO: 103930

Lat: 52.2086N

Lon: 14.1181E

Elev: 112

StdP: 99.99

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.1	-10.4	-16.3	0.9	-11.6	-13.2	1.2	-9.1	12.4	6.2	10.9	6.3	2.5	40	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.7	19.0	27.7	18.3	26.0	17.6	20.2	27.4	19.3	26.0	18.4	24.4	2.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.0	22.1	16.8	12.2	21.4	16.0	11.5	20.8	58.4	27.6	55.3	26.3	52.4	24.3	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.3	7.2	-13.8	33.9	5.0	2.0	-17.4	35.4	-20.3	36.5	-23.1	37.7	-26.8	39.1		
(5)			-14.1	22.2	4.9	0.7	-17.7	22.8	-20.5	23.2	-23.3	23.6	-26.9	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-0.2	-0.1	4.3	8.1	13.6	16.3	18.8	18.5	14.4	9.7	4.0	1.2	(6)
(7)		DBStd	7.92	5.67	4.99	3.83	3.76	3.92	3.58	3.41	3.50	2.97	3.97	3.58	4.00	(7)
(8)		HDD10.0	1381	317	282	179	81	13	1	0	0	1	54	180	272	(8)
(9)		HDD18.3	3499	575	515	435	306	154	83	37	42	124	269	429	530	(9)
(10)		CDD10.0	1052	0	1	3	24	123	188	272	263	133	44	1	1	(10)
(11)		CDD18.3	131	0	0	0	0	6	21	51	47	6	1	0	0	(11)
(12)		CDH23.3	1137	0	0	0	2	57	191	448	394	42	2	0	0	(12)
(13)		CDH26.7	297	0	0	0	0	3	41	127	122	4	0	0	0	(13)
(14)	Wind	WSAvg	3.2	4.0	3.5	3.7	3.1	2.7	2.8	2.8	2.6	2.9	3.2	3.5	3.8	(14)
(15)	Precipitation	PrecAvg	553	40	33	38	36	54	61	66	60	43	40	43	45	(15)
(16)		PrecMax	793	101	85	101	93	171	154	194	214	141	126	109	108	(16)
(17)		PrecMin	357	0	1	8	1	10	7	6	12	5	4	1	2	(17)
(18)		PrecStd	111	19	18	21	20	33	34	41	38	26	26	21	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	14.3	17.4	22.5	26.8	30.6	32.2	33.8	27.1	22.5	13.7	11.6	(19)
(20)			MCWB	8.6	10.0	10.7	13.5	16.9	19.0	20.3	20.6	18.1	16.2	10.7	9.1	(20)
(21)		2%	DB	8.9	10.8	13.9	19.4	25.0	27.6	29.7	29.7	24.4	19.2	11.6	10.3	(21)
(22)			MCWB	7.1	8.0	9.2	12.1	15.7	17.8	19.0	19.3	17.0	14.5	9.4	8.4	(22)
(23)		5%	DB	7.6	8.0	11.6	17.2	23.1	25.6	27.9	27.4	22.0	17.0	10.3	8.6	(23)
(24)			MCWB	6.0	6.0	7.6	10.9	14.8	17.0	18.4	18.5	15.9	13.1	8.5	7.1	(24)
(25)		10%	DB	6.3	6.0	9.8	14.8	20.9	23.4	25.9	25.2	19.6	15.3	8.9	7.2	(25)
(26)			MCWB	4.9	4.3	6.9	9.7	13.8	16.1	17.8	17.7	14.8	12.0	7.4	6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.3	11.6	14.1	18.5	20.6	21.8	21.2	19.4	17.3	11.2	9.6	(27)
(28)			MCDWB	10.3	13.4	15.9	20.6	24.9	26.4	29.0	30.6	25.4	21.2	12.8	11.1	(28)
(29)		2%	WB	7.5	8.0	9.6	12.8	16.8	19.2	20.2	20.1	17.7	15.2	9.7	8.7	(29)
(30)			MCDWB	8.9	10.1	13.0	18.3	22.4	25.5	27.2	27.9	22.4	18.3	11.3	10.2	(30)
(31)		5%	WB	6.2	6.2	8.4	11.4	15.5	17.9	19.3	19.2	16.6	13.6	8.6	7.3	(31)
(32)			MCDWB	7.3	7.9	10.9	16.2	21.1	23.3	26.0	26.0	20.5	16.4	10.0	8.5	(32)
(33)		10%	WB	5.0	4.4	7.1	10.1	14.4	16.7	18.3	18.2	15.5	12.4	7.6	6.0	(33)
(34)			MCDWB	6.2	5.8	9.6	14.5	20.1	21.8	24.6	23.9	19.1	14.8	8.8	7.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	6.1	7.2	9.0	9.8	9.3	9.9	9.9	8.5	7.5	5.2	4.2	(35)
(36)			MCDBR	5.2	8.5	10.3	12.3	12.7	13.0	13.3	13.3	11.9	10.0	6.7	5.4	(36)
(37)			MCWBR	4.4	6.0	6.3	6.7	6.2	5.9	5.9	5.9	6.1	6.0	5.2	4.6	(37)
(38)		5% WB	MCDWR	5.1	8.1	9.5	11.3	11.4	11.1	11.7	12.0	10.4	9.1	6.5	5.3	(38)
(39)			MCWBR	4.5	5.9	6.2	6.5	6.1	5.6	5.9	5.8	5.8	5.9	5.3	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.350	0.390	0.412	0.410	0.407	0.427	0.421	0.383	0.373	0.348	0.317	(40)	
(41)		taud	2.314	2.291	2.199	2.217	2.266	2.300	2.253	2.285	2.367	2.348	2.353	2.332	(41)	
(42)		Ebn at Noon	662	751	788	820	842	848	821	804	794	715	625	602	(42)	
(43)		Edh at Noon	64	86	114	127	127	124	128	118	97	81	62	54	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.67	1.36	2.41	4.01	4.91	5.22	4.91	4.26	3.09	1.73	0.79	0.49	(44)	
(45)		RadStd	0.09	0.24	0.30	0.54	0.58	0.42	0.62	0.38	0.38	0.22	0.11	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page